

ABINU ALFRED

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OBJECTIVE

Motivated B.Tech student in Robotics and AI with hands-on experience in embedded systems, simulation, and IoT. Possessing foundational knowledge of ROS 2 and a strong drive to deepen this expertise, I am seeking an internship at Genrobotics to contribute to innovative robotic solutions while mastering ROS-driven development.

EDUCATION

- Bachelor of Technology, Robotics and AI – (2024 - Present)
Rajiv Gandhi Institute of Technology, Kottayam | Ongoing
CGPA: 8.27
- Higher Secondary Education (HSE) - Science Stream (2021 - 2023)
St. Joseph HSS Mathilakam | Completed with 97.9%
- Secondary School Leaving Certificate (SSLC) - (2021)
St. Joseph HSS Mathilakam | Completed with Full A+

SKILLS

- **Technical Skills:** C, Python, SQL (Basic)
- **Web Development:** HTML, CSS, JavaScript (Basic)
- **Core Areas:** AI, Computer Vision, IoT, Embedded Systems
- **Tools & Platforms:** ROS2 (Basic), Git & GitHub, Arduino IDE, Linux, VS Code, ArduPilot SITL, Gazebo, QGroundControl
- **Engineering Skills:** Hardware Prototyping, Robotics System Design, Sensor Integration, Simulation & Testing
- **Soft Skills:** Problem-Solving, Creative Thinking, Time Management, Teamwork, Leadership, Effective Communication, Project Management

PROJECTS

- **Disaster Management Drone Simulation** – Engineered and simulated quadcopter and custom VTOL drones for disaster management using ArduPilot SITL, Gazebo, and QGroundControl; worked on a non-standard VTOL design and validation as part of a 10-member team in a national-level competition.
- **HerWell Women's Health Website** – Developed HERWELL, a women's health web platform enabling period tracking, symptom logging, personalized health insights, and diet recommendations; achieved **3rd position** in a national-level hackathon (IGNIFEX) as part of a 4-member team.
- **Smart Environmental Monitor (IoT Project)** – Developed a Smart Environmental Monitoring System using Arduino, ESP8266, and sensors to measure temperature, humidity, light intensity, and air quality; designed a real-time visualization interface and collaborated in a team for a technical hackathon.
- **Runo – Smart Runaway Alarm Clock** - Developed Runo, a smart runaway alarm clock using ESP32, RTC module, and motor drivers that prevent snoozing through movement; implemented obstacle-aware navigation and customizable alarms as part of a hardware innovation challenge.

LEADERSHIP EXPERIENCE

- **Technical Coordinator, IEEE RAS Student Branch RIT (2026 - Present)**

Coordinated technical events and managed communications and outreach, enhancing student engagement.

- **Technical Lead, ISTE Mechatronics Society Student Chapter RIT (2025 - 2026)**

Led technical initiatives, guided team members in projects, and coordinated club activities

- **NSS Leader, National Service Scheme (Higher Secondary Level) (2021 - 2023)**

Led volunteer teams in social service activities, organized community programs, and camps

CERTIFICATIONS

- NIDAR | Drone Federation of India | Team Member | 2025
- Cybersecurity 101: Navigating Cyberspace Safely | IEEE | 2025
- Data Science with Python Workshop | Techmaghi & IIT Hyderabad | 2024
- Computer Vision Workshop (REX 5.0) | IEEE | 2024